Notes: Bartelme, Costinot, Donaldson, and Rodríguez-Clare (JPE, 2025)

The Textbook Case for Industrial Policy: Theory Meets Data

Contents

1 Big picture

- **Question.** How large are the welfare gains from the *textbook* case for industrial policy, that is, subsidies targeted at sectors with external economies of scale?
- **Setting.** The paper studies a multisector open economy in which sector-level external economies of scale create a wedge between private and social marginal cost. It then brings this framework to data using world trade flows.
- **Core challenge.** The usual theoretical case for industrial policy is clear, but the relevant object—sector-level scale elasticity—is hard to estimate credibly. One needs a way to identify how much larger sector size lowers costs using data that are actually available across countries.
- **Main theoretical result.** In the baseline environment, optimal industrial policy takes the form of sector-specific employment subsidies, while optimal trade policy takes the form of export taxes. The gains from policy can be approximated by Harberger triangles whose *height* is the wedge and whose *base* is the reallocation induced by policy.
- **Main empirical result.** Using bilateral trade data and external estimates of trade elasticities, the paper finds substantial external economies of scale across manufacturing sectors, with an average scale elasticity around 0.21, statistically significant in every manufacturing sector.
- **Main quantitative result.** Despite those sizable scale elasticities, gains from optimal industrial policy are modest in the benchmark calibration: the unweighted average gain is about 1.08% of initial income. In a richer environment with physical capital and input-output linkages, the average rises to 4.06%, but the paper still concludes that the textbook case is not transformative.
- **Central intuition.** The wedges are not tiny. The problem is that optimal policy generates only modest reallocations across sectors. In the paper's language, the Harberger triangles have meaningful heights but small bases.
- **Why this paper matters.** It gives a clean bridge from industrial-policy theory to quantitative evidence. The paper shows how to estimate scale elasticities directly from trade data and how to convert them into welfare calculations in general equilibrium.

2 Data and measurement (key objects)

2.1 Bilateral trade data and country-sector coverage

- **Trade flows.** The core data are bilateral trade flows $X_{ij,k}$ from the OECD Inter-Country Input-Output (ICIO) tables.
- **Country coverage.** The ICIO sample contains 61 countries.
- **Sector coverage for estimation.** The paper estimates scale elasticities for 15 manufacturing sectors. The calibration then adds 17 nonmanufacturing sectors.
- **Domestic sales.** The ICIO tables include domestic transactions $X_{ii,k}$, which lets the authors construct total expenditure and total sales by country and sector.
- **Baseline year.** The main estimates use the 2010 cross section, with robustness checks in 1995, 2000, and 2005 and in a pooled 1995–2010 specification.

2.2 Population, sector size, and sales shares

- **Population.** Country labor supply is proxied by population $\hat{L}_i$ from the Penn World Tables.
- **Sector sales.** Let $S_{i,k} = \sum_j X_{ij,k}$ denote sales by country i in sector k, and $S_i = \sum_{j,k} X_{ij,k}$ total sales.
- **Sector size.** In the baseline labor-only model, the wage bill of sector k equals total sales, so sector employment can be constructed as

$$L_{i,k} = \left(\frac{S_{i,k}}{S_i} \right) \hat{L}_i.$$

- **Interpretation.** A sector is "large" when it absorbs a large share of a country's labor supply.

2.3 Auxiliary elasticities used in estimation and calibration

- **Trade elasticities.** The paper imports sector-level trade elasticities v_k from Giri, Yi, and Yilmazkuday (2021). These are used both in estimation and in the policy counterfactuals.
- **Sectoral substitution.** The paper also estimates the elasticity of substitution across sectors. The IV estimate is $\hat{r} = -0.13$, so the implied elasticity $1 + \hat{r}$ is about 0.87, close to Cobb–Douglas and in the complements range.
- **Nonmanufacturing calibration.** For nonmanufacturing sectors, the calibration sets $g_k = 0$ and assigns $v_k = 4.32$, the median trade elasticity across manufacturing sectors.

2.4 Unobservables and residual objects

- **Productivity.** $A_{ij,k}$ captures origin-destination productivity and trade costs.
- **Demand shifters.** Country-sector tastes enter through $b_{j,k}$.
- **Residual productivity term.** The estimating equation works with a demeaned residual $e_{i,k}$, which captures unobserved country-sector productivity after removing country and sector fixed effects.
- **Demand-predicted size.** Instrumental variables are built from a demand-based prediction of sector size, denoted $\hat{L}_{i,k}$.

2.5 Main measured outputs of the empirical exercise

The empirical analysis ultimately produces four key objects:

- sector-level scale elasticities g_k,
- optimal employment subsidies and trade taxes,
- exact welfare gains from optimal industrial and trade policy,
- and second-order Harberger-triangle approximations to those gains.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline environment: countries, sectors, and technology

Consider countries indexed by i or j, and sectors indexed by k. Labor is the only factor in the baseline model, with fixed national labor supply L_i.

Production for goods from origin i, sector k, sold to destination j is

$$y_{ij,k} = A_{ij,k} E_k(L_{i,k}) \ell_{ij,k},$$

where $\ell_{ij,k}$ is labor used to produce that shipment and

$$L_{i,k} = \sum_j \ell_{ij,k}$$

is total employment in country i, sector k.

- **Interpretation.** Holding sector size fixed, firm-level production is constant returns to scale. External economies are captured by $E_k(L_{i,k})$: larger sector size lowers effective unit cost.
- **Why this matters.** If $E_k(\cdot)$ rises with sector size, individual firms ignore part of the social benefit of expanding activity in high-scale sectors.

3.2 Preferences and demand

Each country j has a representative consumer with utility $u_j(c_j)$, where c_j collects consumption across origins and sectors. In the empirical section, the paper imposes nested CES preferences:

$$u_j = \left(\sum_k b_{j,k}^{1/(1+r)} C_{j,k}^{r/(1+r)} \right)^{(1+r)/r},$$

with sectoral subaggregator

$$C_{j,k} = \left(\sum_i c_{ij,k}^{v_k/(1+v_k)} \right)^{(1+v_k)/v_k}.$$

- **Interpretation of $1 + v_k$.** It is the elasticity of substitution between goods from different origins within sector k.
- **Interpretation of $1 + r$.** It is the elasticity of substitution between sectors.
- **Why nested CES is useful.** It yields gravity equations for trade and a tractable mapping from trade flows to prices and costs.

3.3 The government's problem and the two distortions

Country j's government chooses employment subsidies $s_{j,k}$, import taxes $t^m_{ij,k}$, and export taxes $t^x_{ji,k}$ to maximize the utility of its representative consumer.

A small policy change implies the envelope condition

$$\frac{dU_j}{\lambda_j} = -\sum_k s_{j,k} w_j dL_{j,k} + \sum_{i\neq j,k} t^m_{ij,k} \tilde{p}_{ij,k} dc_{ij,k} + \sum_{i\neq j,k} t^x_{ji,k} \tilde{p}_{ji,k} dy_{ji,k} \tag{1}$$

$$+ \sum_k e^E_{j,k}(1 - s_{j,k}) w_j dL_{j,k} + \sum_{i\neq j,k} e^p_{ji,k} \tilde{p}_{ji,k} dy_{ji,k}, \tag{2}$$

where

$$e^E_{j,k} = \frac{d\ln E_k(L_{j,k})}{d\ln L_{j,k}}$$

is the scale elasticity and $e^p_{ji,k} = d\ln \tilde{p}_{ji,k}(y_{ji,k})/d\ln y_{ji,k}$ is the export-price elasticity.

- **Interpretation.** The first line is the marginal deadweight loss from taxes and subsidies. The second line is the marginal benefit from correcting the two distortions: external economies of scale and market power in export markets.
- **Economic message.** Industrial policy corrects production-side externalities; trade policy corrects terms-of-trade distortions.

3.4 Optimal industrial and trade policy

In the baseline environment, the paper shows that optimal policy has the form

$$s^*_{j,k} = \frac{\bar{s}_j + e^{E*}_{j,k}}{1 + e^{E*}_{j,k}}, \qquad t^{m*}_{ij,k} = \bar{t}_j, \qquad t^{x*}_{ji,k} = 1 - (1 + \bar{t}_j)(1 + e^{p*}_{ij,k}),$$

for some shifts $\bar{s}_j$ and $\bar{t}_j$, which are normalized to zero in the quantitative analysis.

A convenient intuitive special case is

$$s^*_{j,k} = \frac{e^{E*}_{j,k}}{1 + e^{E*}_{j,k}}, \qquad t^{m*}_{ij,k} = 0, \qquad t^{x*}_{ji,k} = -e^{p*}_{ji,k}.$$

- **Interpretation.** Sectors with larger scale elasticities get larger employment subsidies.
- **Trade-policy side.** Import tariffs are unnecessary in the small-country benchmark, while export taxes target endogenous export-price distortions.
- **Main intuition.** Pigouvian industrial policy and optimal trade policy are complements in the full optimum.

3.5 Harberger-triangle formulas for welfare gains

A central result of the paper is that policy gains admit a second-order Harberger-triangle representation. For industrial policy,

$$\frac{W^I_j}{I_j} \simeq \frac{1}{2} \sum_k \frac{w_j L_{j,k}}{I_j} \left(\frac{(\Delta L_{j,k})^I}{L_{j,k}} \right) e^{E*}_{j,k}.$$

For trade policy,

$$\frac{W_j^T}{I_j} \simeq \frac{1}{2} \sum_{i \neq j,k} \frac{\tilde{p}_{ji,k} y_{ji,k}}{I_j} \left(\frac{(\Delta y_{ji,k})^T}{y_{ji,k}} \right) e^{p*}_{ji,k}.$$

- **Interpretation.** These expressions are weighted sums of Harberger triangles.
- **Height.** The distortion itself, either the scale elasticity or export-price elasticity.
- **Base.** The induced reallocation in employment or exports.
- **Immediate implication.** Large wedges are not enough. Big welfare gains also require big reallocations.

3.6 Nonparametric idea for identifying scale elasticities

The paper's core empirical idea is simple:

- if a sector has positive external economies of scale, larger sectors should have lower prices;
- if prices are lower, expenditure and quantities demanded should be higher.

In principle, one could identify $E_k(\cdot)$ nonparametrically from demand-driven changes in sector size. In practice, with only 61 countries, the paper proceeds parametrically.

3.7 Parametric assumptions and the baseline estimating equation

The paper assumes constant scale elasticities within sectors:

$$E_k(L_{i,k}) = L_{i,k}^{g_k}.$$

Under the nested CES structure, the main estimating equation becomes

$$Y_{i,k} = d_i + d_k + g_k \ln L_{i,k} + e_{i,k},$$

where

$$Y_{i,k} \equiv \frac{1}{v_k} \left(\frac{1}{J} \sum_j \ln X_{ij,k} \right),$$

and d_i and d_k are country and sector fixed effects.

- **Interpretation.** Once trade flows are adjusted by the trade elasticity, a country-sector with larger scale should systematically export more if larger scale reduces cost.
- **What g_k means.** It is the elasticity of productivity with respect to sector size. A value of 0.21 means a 1 percent increase in sector size raises productivity by about 0.21 percent.

3.8 Trade elasticities and gravity

Nested CES demand implies the standard gravity relationship

$$\ln \left(\frac{X_{ij,k}}{X_{j,k}} \right) = -v_k \ln \left(\frac{p_{ij,k}}{P_{j,k}} \right).$$

The paper does not reestimate v_k; instead, it imports sector-level trade elasticities from Giri, Yi, and Yilmazkuday (2021).

- **Why this is important.** The same export response can imply a small or large scale elasticity depending on how price-sensitive demand is in that sector.
- **Example.** Similar reduced-form responses in paper and basic metals translate into very different scale elasticities because paper has much lower trade elasticity than metals.

3.9 Demand-side instruments for sector size

The main identification problem is that $L_{i,k}$ is endogenous. The paper therefore constructs demand-predicted sector size using country population and demand shifters.

A convenient representation is

$$\hat{b}_{i,k} = \frac{x_{i,k}(\hat{P}_{i,k})^{\hat{r}}}{\sum_l x_{i,l}(\hat{P}_{i,l})^{\hat{r}}}, \qquad \hat{L}_{i,k} = \hat{b}_{i,k}\hat{L}_i.$$

These demand-based predictions are then interacted with sector indicators and used as instruments for $\ln L_{i,k}$.

The exclusion restriction is

$$\mathbb{E}\left[\ln \hat{L}_{i,k}\, e_{i,k} \mid k\right] = 0 \qquad \text{for all } k.$$

- **Interpretation.** Variation in predicted sector size must come from demand-side forces, not from sector-specific productivity advantages.
- **Potential threats.** Larger countries might have sector-specific technological advantages; or countries with stronger tastes for a sector might also be especially good at producing in it.

3.10 Estimating the sectoral substitution elasticity r

The upper-level substitution elasticity is itself estimated by IV from cross-country expenditure patterns and price proxies. The paper finds

$$\hat{r} = -0.13,$$

so the implied elasticity of substitution across sectors is

$$1 + \hat{r} = 0.87.$$

- **Interpretation.** Sectors are not very substitutable in the data.
- **Quantitative implication.** Even when wedges are meaningful, policy-induced reallocations across sectors will be limited.

3.11 Calibration and counterfactual analysis

The calibrated world economy contains:

- 61 countries,
- 15 manufacturing sectors with estimated g_k,
- 17 nonmanufacturing sectors with $g_k = 0$,
- sector-level trade elasticities v_k,
- and upper-level substitution parameter $r = -0.13$.

The paper then solves the nonlinear counterfactual system to compute exact gains from:

- fully optimal policy,
- optimal trade policy,
- and optimal industrial policy.

3.12 Beyond Ricardian economies: capital and input-output linkages

The richer environment adds physical capital and intermediate input linkages:

$$y_{ij,k} = A_{ij,k} E_k(Z_{i,k}) z_{ij,k}, \qquad z_{ij,k} = f_k(\ell_{ij,k}, k_{ij,k}, m_{ij,k}),$$

with composite input size

$$Z_{i,k} = \sum_j z_{ij,k}.$$

Optimal policy remains Pigouvian in spirit. In this environment,

$$s^*_{j,k} = \frac{e^{E*}_{j,k}}{1 + e^{E*}_{j,k}}, \qquad t^{m*}_{ij,k} = \bar{t}_j, \qquad t^{x*}_{ji,k} = 1 - (1 + \bar{t}_j)(1 + e^{p*}_{ij,k}).$$

The estimating equation becomes

$$Y^{IO}_{i,k} = d_i + d_k + g_k \ln Z_{i,k} + e_{i,k},$$

where $Y^{IO}_{i,k}$ additionally controls for relative factor prices and intermediate-input prices.

- **Economic point.** Gross flows matter more than value added in this environment, so the effective tax base is larger.
- **Quantitative point.** This is why gains rise substantially in the extended model.

4 Identification and instruments (what makes the estimates credible)

4.1 Why naive OLS is not enough

- Sector size $L_{i,k}$ is endogenous because countries specialize where they have high productivity.
- Therefore OLS on $Y_{i,k} = d_i + d_k + g_k \ln L_{i,k} + e_{i,k}$ confounds external economies of scale with comparative advantage.
- The paper indeed finds that OLS estimates are generally a bit larger than IV estimates, which is exactly what one expects if high-productivity sectors are also larger.

4.2 What identifies g_k in the new strategy

- The paper seeks *demand-driven* variation in sector size.
- The instrument combines population and demand residuals to predict how large a country-sector should be because of demand, not because of supply.
- The logic is a home-market-effect logic: if a country has unusually strong demand for a sector, that should expand the sector, and with increasing returns it should also raise exports.

4.3 Why the trade elasticities are central but not decisive for welfare

- Trade elasticities v_k are central for recovering g_k from the trade data.
- They also affect counterfactual reallocations because more elastic foreign demand allows bigger output and employment shifts.
- However, the paper shows that while the exact ranking of sectors by subsidy can be sensitive to v_k, the aggregate welfare gains are much less sensitive than one might think.

4.4 Main exclusion restriction and its threats

The key exclusion restriction is

$$\mathbb{E}\left[\ln \hat{L}_{i,k} e_{i,k} \mid k\right] = 0.$$

The paper discusses two main threats:

- **Country-size threat.** Larger countries may be intrinsically more productive in some sectors.
- **Taste-technology correlation threat.** Countries with stronger tastes for some sectors may also have unobserved supply-side advantages in those sectors.

The paper probes these threats by:

- including controls for exporter per-capita GDP interacted with sectors,
- checking robustness to alternative years and pooled samples,
- checking robustness to alternative values of r,
- and showing that estimates of g_k are extremely stable across these changes.

4.5 Why the first stage is credible

- The first stage is sector-specific: demand-predicted size strongly explains own-sector size and only weakly explains other sectors' sizes.
- Both the conventional and Sanderson–Windmeijer first-stage statistics are large.
- This is important because the paper estimates 15 sector-specific scale elasticities at once.

4.6 Relation to the literature

- Relative to the gravity literature, the key contribution is to estimate scale elasticities *directly* rather than back them out indirectly from monopolistic-competition calibrations.
- Relative to industrial-policy theory, the paper is the bridge from Pigouvian logic to calibrated numbers.
- Relative to urban and regional economics, the paper works at the country-sector level rather than the city or cluster level.

5 Tables and figures

5.1 Figure 1: Textbook Pigouvian industrial policy

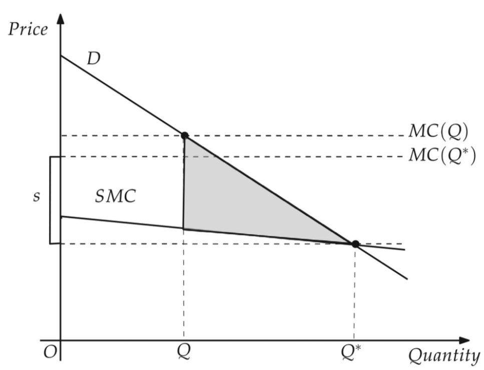

FIG. 1.—Textbook case for industrial policy. Because of external economies of scale in a sector, private marginal cost MC exceeds social marginal cost SMC, and so output Q is less than the social optimum Q^*. The optimal industrial policy is a subsidy $s = \mathrm{MC}(Q^*) - \mathrm{SMC}(Q^*)$, which gives rise to gains equal to the area of the gray Harberger triangle.

TABLE 1
ESTIMATES OF SECTOR-LEVEL SCALE ELASTICITIES (γ_k)

				First Stage		
Sector	OLS (1)	IV (2)	Reduced Form (3)	F (4)	SW F (5)	Trade Elasticity (6)
Food, beverages, and tobacco	.26	.24	.28	192.4	1,678.9	3.57
	(.02)	(.02)	(.03)			(.27)
Textiles	.22	.21	.24	117.9	2,216.8	4.43
	(.02)	(.02)	(.03)			(.31)
Wood products	.20	.19	.22	218.8	1,383.6	4.32
	(.04)	(.04)	(.04)			(1.03)
Paper products	.32	.31	.35	138.1	1,608.7	2.97
	(.05)	(.05)	(.05)			(.42)
Coke/petroleum products	.10	.11	.15	14.8	472.8	8.94
	(.05)	(.05)	(.06)			(5.33)
Chemicals	.23	.22	.26	70.1	1,192.6	3.75
	(.03)	(.03)	(.04)			(.51)
Rubber and plastics	.21	.20	.23	154.9	1,595.4	4.13
	(.05)	(.05)	(.05)			(1.18)
Mineral products	.20	.19	.22	301.6	1,485.0	5.14
	(.04)	(.03)	(.04)			(1.17)
Basic metals	.12	.11	.13	137.1	1,342.2	8.94
	(.05)	(.05)	(.06)			(5.33)
Fabricated metals	.19	.18	.21	191.1	2,321.4	5.07
	(.08)	(.09)	(.09)			(2.46)
Machinery and equipment	.28	.26	.31	136.8	1,764.6	3.27
	(.05)	(.04)	(.05)			(.52)
Computers and electronics	.28	.27	.32	71.8	612.5	3.27
	(.04)	(.04)	(.05)			(.52)
Electrical machinery (National	.27	.25	.30	109.7	1,474.2	3.27
Electrical Code)	(.04)	(.04)	(.05)			(.52)
Motor vehicles	.20	.19	.26	80.4	1,392.2	4.47
	(.04)	(.04)	(.04)			(.78)
Other transport equipment	.20	.20	.24	25.7	846.8	4.47
	(.03)	(.04)	(.04)			(.78)

NOTE.—Columns 1 and 2 report the OLS and IV estimates, respectively, of γ_k in eq. (25). Column 3 reports the reduced form coefficients. The instruments are the log of (country population × sectoral demand shifter), interacted with sector indicators. All regressions in cols. 1–3 include exporter and sector fixed effects. Columns 4 and 5 report the conventional and Sanderson-Windmeijer (SW) F-statistics, respectively, from the first-stage regression corresponding to the endogenous regressor formed by interacting the log of sector size with an indicator for the sector named in each row. Column 6 reports the estimate of θ_k from Giri, Yi, and Yilmazkuday (2021) used to construct $Y_{i,k}$ in eq. (25). Standard errors in cols. 1–3 are clustered at the exporter level and adjusted to account for the uncertainty in the estimates of θ_k, as described in app. sec. B.5. Standard errors are in parentheses.

Read:

- The figure is the paper's organizing picture. External economies of scale make private marginal cost exceed social marginal cost.
- The resulting underproduction creates a Pigouvian rationale for a subsidy equal to the gap between private and social marginal cost at the efficient quantity.
- The welfare gain is the shaded Harberger triangle.

- The whole paper is about estimating how tall and how wide those triangles are in actual data.

5.2 Table 1: Estimated scale elasticities across manufacturing sectors

Read:

- The preferred IV estimates imply statistically significant external economies of scale in every one of the 15 manufacturing sectors.
- The average scale elasticity is about 0.21.
- There is important heterogeneity: the low end is about 0.11 in coke/petroleum products and basic metals, while the high end is about 0.31 in paper products.
- Other sectors lie in the middle: food and beverages about 0.24, chemicals about 0.22, machinery and equipment about 0.26, computers and electronics about 0.27, and motor vehicles about 0.19.
- The first-stage statistics are strong, which is important because the entire quantitative exercise rests on these estimates.

5.3 Figure 2 and Table 2: Exact gains and the Harberger-triangle approximation

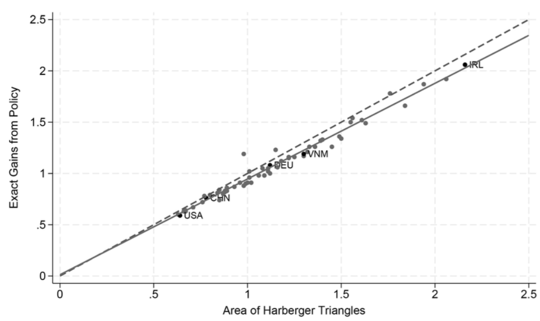

FIG. 2.—Exact policy gains versus areas of Harberger triangles. The figure reports the exact gains from industrial policy (as a share of initial income) on the *y*-axis, as defined in equation (14), against the second-order approximation from proposition 2 on the *x*-axis, as described in equation (17). Solid line = line of best fit; dashed line = 45° line.

TABLE 2
GAINS FROM OPTIMAL POLICIES: SELECTED COUNTRIES

Country	Fully Optimal (1)	Trade (2)	Industrial (3)
United States	.82	.51	.59
China	.91	.74	.76
Germany	1.35	1.51	1.08
Ireland	2.72	2.35	2.06
Vietnam	2.07	1.88	1.19
Average:			
Unweighted	1.67	1.46	1.08
GDP weighted	1.06	.87	.76

NOTE.—Column 1 reports the gains associated with fully optimal policies, as defined in eq. (13). Column 2 reports the gains associated with optimal trade policy, as defined in eq. (15). Column 3 reports the gains associated with optimal industrial policy, as defined in eq. (14). All gains are reported as a share (%) of initial income.

Read:

- Figure 2 shows that the exact gains from industrial policy line up tightly with the second-order Harberger-triangle approximation; the slope is about 0.93 and the R^2 is 0.97.
- Table 2 reports the benchmark policy gains. The average gains from optimal industrial policy are 1.08% of income on an unweighted basis and 0.76% on a GDP-weighted basis.
- Selected countries illustrate the heterogeneity: the United States gains 0.59%, China 0.76%, Germany 1.08%, Ireland 2.06%, and Vietnam 1.19%.
- The gains from industrial policy are smaller than those from optimal trade policy on average, and fully optimal policy combines the two.

5.4 Why Table 2 is the paper's core quantitative result

Read:

- The important contrast is between the benchmark gain of 1.08% and the back-of-the-envelope gain of 12.4% that would arise if labor fully reallocated to the sector with the largest scale elasticity.
- The paper's message is therefore not that wedges are small. It is that equilibrium reallocations are limited.
- This is the central "large heights, small bases" insight.

5.5 Figure 3: Industrial policy gains versus openness

Read:

- Gains from industrial policy are strongly increasing in openness, measured as exports plus imports over gross output.
- More open economies can reallocate labor away from shrinking sectors more easily, so the Harberger triangles have larger bases.
- This channel explains much of the cross-country heterogeneity in gains.
- The paper emphasizes that Ireland's much larger gains relative to the United States are mainly an openness result, not a wedge result.

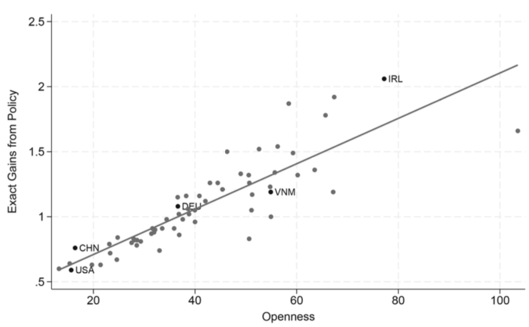

Fig. 3.—Industrial policy gains versus openness. The figure reports the exact gains from industrial policy (as a share of initial income) on the *y*-axis, as defined in equation (14), against openness on the *x*-axis, measured as exports plus imports over gross output.

TABLE 3
Gains from Constrained and Globally Efficient Industrial Policies: Selected Countries

Country	Baseline (1)	Constrained (2)	Globally Efficient (3)
United States	.59	.48	.62
China	.76	.49	.02
Germany	1.08	.60	−.63
Ireland	2.06	1.64	−2.14
Vietnam	1.19	1.28	1.66
Average:			
Unweighted	1.08	.97	.35
GDP weighted	.76	.59	.23

Note.—Each column reports the gains—expressed as a share (%) of initial real national income—that could be achieved by each type of policy.

5.6 Table 3: Industrial policy with trade-policy constraints or global coordination

Read:

- If countries are forced to set trade taxes to zero and can only use industrial policy, the average gain falls slightly from 1.08% to 0.97%.
- If employment subsidies are chosen in a Pareto-efficient coordinated way, the GDP-weighted average gross gain is only 0.23%.

- In that coordinated case, Vietnam still gains 1.66%, but Ireland can lose 2.14% gross of transfers because of adverse terms-of-trade effects.
- This table shows that the gains from industrial policy are very sensitive to whether countries can also manipulate terms of trade and to whether policy is unilateral or coordinated.

5.7 Figure 4 and Figure 5: Sensitivity to trade elasticities and sectoral substitution

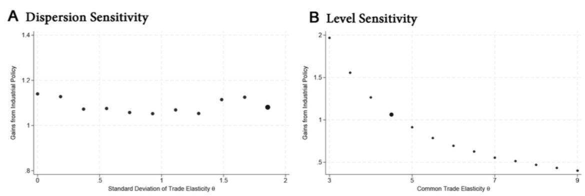

FIG. 4.—Gains from industrial policy: alternative trade elasticities. *A* reports the (global, unweighted average) gains from industrial policy, as defined in equation (14), for different values of the standard deviation of trade elasticities achieved by varying $\lambda \in [0, 1]$ in $\theta_k = \lambda\theta_{k,\text{baseline}} + (1 - \lambda)\text{median}\{\theta_{s,\text{baseline}}\}$. Large circle indicates the trade elasticities used in table 2, that is, $\lambda = 1$. *B* does the same for different values of a common trade elasticity $\theta_k = \theta$. Large circle indicates the median of the estimates from Giri, Yi, and Yilmazkuday (2021), that is, $\theta = 4.32$.

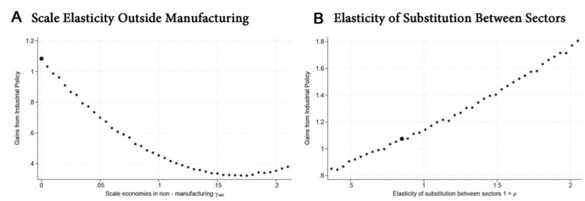

FIG. 5.—Gains from industrial policy: other parameter values. *A* reports the (global, unweighted average) gains from industrial policy, as defined in equation (14), for different values of the scale elasticity outside manufacturing, γ_{NM}. *B* does the same for different values of the elasticity of substitution between sectors, $1 + \rho$. Large circles indicate baseline parameter values used in table 2.

Read:

- Figure 4A shows that changing the dispersion of trade elasticities across sectors has very little effect on the average gains from industrial policy.
- Figure 4B shows that the *level* of trade elasticities matters more: with a common trade elasticity as low as 2.97, average gains rise to roughly 2%.
- Figure 5 shows that even if sectors become substantially more substitutable, gains remain moderate. For an elasticity of substitution across sectors of 2, average gains are still only about 1.76%.
- So the qualitative conclusion—textbook industrial policy is modest on average—is robust to a broad range of calibrated parameters.

5.8 Figure 6 and Figure 7: Beyond the Ricardian baseline

Read:

- Figure 6 shows that the optimal industrial subsidies in the richer model with capital and input-output linkages are very close to those in the Ricardian baseline.
- Figure 7 shows that the *gains* are much larger in the richer environment.
- The average industrial-policy gain rises from 1.08% in the baseline to 4.06% with capital and input-output linkages.
- Most of this increase comes mechanically from the fact that policy now acts on gross flows rather than value added, which enlarges the relevant tax base.
- Even so, the gains remain far below the 12.4% full-reallocation back-of-the-envelope benchmark, so the main message survives.

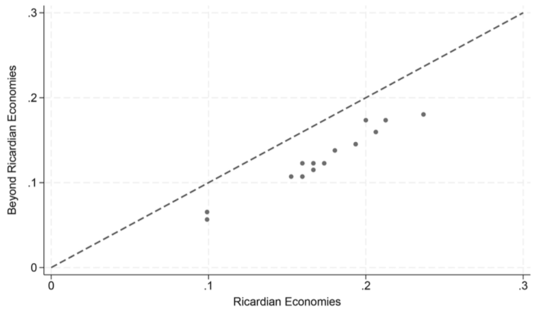

FIG. 6.—Optimal industrial policy beyond Ricardian economies. The y-axis reports the optimal industrial policy $s_{j,k}^{*} = \gamma_k/(1+\gamma_k)$ estimated for 15 manufacturing sectors in the environment with capital and input-output linkages (as described in sec. VI.A), whereas the x-axis reports the optimal industrial policy $s_{j,k}^{*} = \gamma_k/(1+\gamma_k)$ estimated in the baseline environment (as described in sec. II.A).

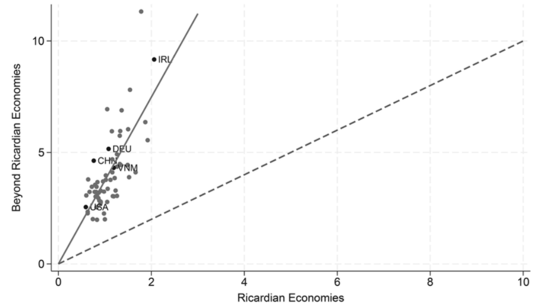

FIG. 7.—Gains from industrial policy beyond Ricardian economies. The figure reports the gains from industrial policy, as defined in equation (14), in the environment with capital and input-output linkages (as described in sec. VI.A) on the y-axis against the gains from industrial policy in the baseline environment (as described in sec. II.A) on the x-axis.

6 Short summary

- **Conclusion.** The paper gives the first quantitative evaluation of the textbook Pigouvian case for industrial policy using estimated sector-level scale elasticities.
- **Identification insight.** Scale elasticities are identified from demand-driven variation in sector size using bilateral trade data, gravity structure, and demand-side instruments.
- **Main empirical lesson.** Manufacturing sectors do display meaningful external economies of scale, but the welfare gains from optimal industrial policy are modest because sectoral reallocation is limited.
- **Quantitative headline.** Average gains from optimal industrial policy are about 1.08% in the baseline and 4.06% in a richer environment with capital and input-output linkages.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that one can estimate sector-level external economies of scale directly from trade data and use those estimates to quantify optimal industrial policy.
- The estimated scale elasticities are economically meaningful, with a manufacturing average near 0.21 and significant variation across sectors.
- Yet the welfare gains from the textbook industrial-policy motive are small to moderate, not transformative.

7.2 Economic intuition

- There are two ingredients in any Harberger-triangle calculation: the wedge and the induced reallocation.
- This paper finds that the wedge—the distance between private and social marginal cost—is not trivial.

- But consumers do not substitute strongly across sectors, and foreign demand is not infinitely elastic. Because of that, labor does not move much from low-scale to high-scale sectors.
- The result is a triangle with noticeable height but a short base.

7.3 Takeaway

The paper is best understood as a disciplined warning against overreading the classical case for industrial policy. It does *not* say that industrial policy can never matter. It says something narrower and more precise: even when sector-level external economies of scale are present and statistically important, the textbook Pigouvian rationale by itself delivers limited gains at the country-sector level that the paper studies. The reason is not that the wedges are tiny, but that equilibrium reallocation is hard. This is exactly the kind of result that makes the paper valuable: it converts a familiar verbal argument into an empirical object, estimates the crucial elasticity directly, and shows how much that argument is actually worth in data.